Detecting Meaningful Change in Injured Workers' Walking with Affordable Sensors

Research summary for faculty review

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Goal

Rehabilitation teams and workers' compensation case managers follow patients through recovery from lower-limb injury. Laboratory force plates can measure how walking changes between visits, but few clinics have them. **My goal is to find out whether affordable sensing, such as a pressure insole, can detect meaningful change in an injured patient's walking, and what it would need to do so.**

Data. I used GaitRec (Horsak et al., *Scientific Data*, 2020), an open dataset from a rehabilitation centre of the Austrian Workers' Compensation Board. The analysis covers 1,667 patients with knee, ankle, hip or heel injuries: 5,870 visits and about 100,000 force-plate steps. A further 92 later patients were kept untouched as a temporal hold-out. I built every measure from the raw force curves.

Study 1: when is a change real?

The usual limit for a "real" change uses same-day noise from repeated steps. With that limit, **28%** of visit-to-visit changes crossed it, where 5% were expected. Measurements move about twice as much between visits as within one.

A corrected limit adds that visit-to-visit variation, estimated on separate patients. It gave 4.7–5.3% in held-out patients, and held in each injury group. The cost is that it misses most small declines.

Study 2: can AI recover what an insole cannot measure?

An insole measures only vertical force. Push-off and braking come from the front–back force, which needs a force plate. I trained models to reconstruct push-off and braking from vertical force: ridge regression, gradient boosting and a 1D CNN. The CNN was written in NumPy, with its gradients checked numerically.

Signal	Agreement with real push-off change (CCC)	Direction correct
Peak vertical loading	0.00	59%
Full vertical curve, gradient boosting	0.39	89%
Full vertical curve, CNN	0.41	89%

So the **shape** of the vertical curve carries push-off information, while its peak does not. But the AI explained only about a quarter of the variation in change.

Study 3: can AI detect change directly?

Under a protocol frozen before analysis, I trained paired-visit models, including a four-channel CNN, to detect real push-off changes directly. Results at a 5% false-positive rate:

Method	Development set, 5-fold CV (243 changes)	Temporal hold-out (12 changes)
Peak vertical loading	9%	8%
AI reconstruction	32%	8%
Paired-visit CNN	31%	8% (95% CI 0–33%)

Detecting change directly did not beat reconstruction.

What this shows. Standard limits mistake ordinary variation for change, and a corrected limit fixes this across four injury groups. The full vertical-force curve carries push-off information that peak loading misses. Across the evaluated models,

though, vertical force could not detect most force-plate-defined push-off changes, and more sophisticated modelling did not resolve this.

What it does not show. That detected changes matter clinically, or that real insoles behave like force-plate vertical signals. Nothing was compared against pain, function or return to work. The hold-out is small. All analysis plans were frozen internally, not publicly registered. No outside reviewer has checked the work yet.

What it means for an affordable device

The results set seven requirements for a workers' compensation rehabilitation tracker. Among them:

- use between-visit-calibrated alert limits;
- confirm declines at a second visit;
- record the full force curve;
- sample at 100 Hz or more;
- compare each leg with its own history and with healthy norms;
- use measurement-only wording, with no MMI, effort or work-capacity language.

The next step is a study testing whether a **portable shear or inertial sensor** closes the push-off gap. That study needs about 270 patients with simultaneous force plates. The proposal is drafted.

Where I would value your guidance

1. **Sensor study.** Is a study of simultaneous force-plate and shear-insole recordings in rehabilitation patients feasible, and what would make it credible?
2. **Clinical meaning.** How should the size of a clinically meaningful decline be set?
3. **Clinical evaluation.** What would a clinician-facing version need before evaluation?

The code (37 tests, with independent re-implementations), three reports, frozen protocols and an interactive prototype are at github.com/sohaarian/gaitsense-clinical-ai (DOI 10.5281/zenodo.23611092).